

STOCK MARKET TASKS

Warm up Tasks

Reference Notes:

I will share some information about domain knowledge.

They key component of the DS is the domain knowledge and understanding the data.

Since we are working on Stock data. I have some reference places to understand data, to get market information, and to make correlations with what we are doing.

Specifically, you can start with the following lectures.

Trends

https://youtu.be/5_NlhAR6_rA

-Adam Khoo explains chart analysis pretty well.

<https://www.youtube.com/channel/UCK-aQjEvZNJI3HINja0gAiQ>

Casino example

<https://youtu.be/bRCtBRsLPmk>

Profitable trading

<https://youtu.be/jwS9yKUplAo>

Popular technical indicators

https://youtu.be/q-fc_TQSL2g

If you can, please also watch about "option trading"

Following pages are also helpful for decision making

Good charts

-Barchart.com

-Finviz.com

Earning information

-earningswhisper.com

Stock news

-marketwatch.com

Social media follow

-stocktwits.com

Charts and fundamentals

-reuters.com/markets/stocks

Some sample tutorials from youtube.

<https://www.youtube.com/watch?v=Bu0kpU-ozaw>

<https://www.youtube.com/watch?v=qLsxbiZgTho>

<https://www.youtube.com/watch?v=2BrpKpWwT2A>

<https://www.youtube.com/watch?v=d2kXmWzfS0w>

<https://www.youtube.com/watch?v=57qAxRV577c>

<https://www.youtube.com/watch?v=fxLsSax7rvY>

https://www.youtube.com/watch?v=ID85lv_R54U

<https://www.youtube.com/watch?v=QIUxPv5PJ0Y>

Actual project

- Stock Market AI based prediction models

The task for Stock Analysis

There are three components

1. Chart Analysis (indicators)
2. Predictions (ML and DL)
3. Combine with sentiment analysis

Chart Analysis

Use Python to retrieve stock market data and analyze price changes.

1. Fetch both **daily** and **hourly** stock market data using a suitable API.
Visualize the data in several types of plots.
2. Plot **long-term stock data** over multiple years.
Create an interactive graph that allows the user to view different time ranges.
3. Start with **Apple (AAPL)** and plot its stock chart using **daily data** over several years.
4. Add moving averages to the stock charts, such as:
 - **SMA20 and SMA50**, or
 - **SMA50 and SMA150**These should be overlaid on the main stock price chart.
5. Create a table for **five stocks**. Use default examples such as **FAANG stocks plus NVIDIA**, but allow the user to change them.

For each stock, calculate the historical probability that the **next day is positive** if the **current day** changes by:

+2%, +3%, +4%, +5%, and +6%.

Example: if **Apple rises 5% today**, what is the historical probability that **the next day closes positive**?

Use **the last 2 years** of historical data.

6. Repeat the same analysis for **negative daily moves**, such as:
-2%, -3%, -4%, and -5%.
7. Repeat both of the above analyses again using **5 years of historical data**.
8. Create a **comparison table** for the five selected stocks.
Include a simple **correlation / collinearity analysis** between them.
9. Plot a **comparison chart** for **two to five stocks** using **daily data for one year**.
Example: compare **Apple** and **Amazon**.
Normalize or rescale prices if needed so they can be compared clearly.
10. Create another table for the same five stocks showing the probability that the **next day is positive** after:
 - **2 consecutive positive days**
 - **3 consecutive positive days**
 - **4 consecutive positive days**
 - **5 consecutive positive days**
 - **6 consecutive positive days**
11. Repeat the same analysis for **consecutive negative days**.
12. Screen **all Nasdaq and NYSE stocks** and identify the **top 10 trending stocks**, ranked by **daily percentage change**.
13. Repeat the screening, but only include stocks with a **market capitalization of \$5 billion or more**.

This may require obtaining market cap data from an API or web scraping source.

PROJECT

PART 1

- 1) Overview of Stock Indicators, building functions to work with AlphaVantage API.
 - a) Part 1 - Data Collection (building API functions) Basic analysis and Visualization.
 - b) Part 2 - Strategy 1 - Finding Most Linear Stocks over 90 days (~60 trading) and Intraday intervals.
 - c) Part 3 - Strategy 2 - Finding basic indicators for the stocks and plotting them.

PART 2

- 2) Predicting Prices and building pipeline for adding new features: (ML)
 - a) Linear Regression
 - b) RF
 - c) SVM
 - d) XGBoost

- e) LSTM
- f) Prophet
- 3) AWS deployment. (**after LR and LSTM**)

PART 3

- 4) Analysis of stock Sentiment using NLP. Adding new features. (NLP)
- 5) Analysis and implementation of other macroeconomic factors as features such as:
 - a) GDP
 - b) Inflation
 - c) Interest Rates
 - d) Etc...
- 6) Building pipeline for feature optimization using as per study (most used technics)
<https://doi.org/10.1186/s40854-022-00441-7> :
 - a) RF
 - b) PCA
 - c) AE
- 7) Using Models from Step 2 again, check-compare results.

PART 4

- 8) Implement Trading Strategies using Predictions and indicators. (Companies main profile is algotrading)
 - a) Developing logic for Algotrading based on our AI predictions.
 - b) Develop our own indicator
- 9) Portfolio optimization using technics to determine optimal stock allocation sizes.
- 10) Bringing it all to full automated bot.
 - a) Connecting to trading platform via API
 - b) Developing trading logic to execute

Advanced Analysis

- ☐ Find the trending stocks in all Nasdaq indices. I will explain the details over the last 60 days. Best linear curve. <https://www.barchart.com/>
- ☐ Find also trending stocks in during the day. Best linear curve. 8am-10:30am if that is linear during that time. — Vas comment : Then Buy and sell before the day.
- ☐ Implement technical indicators such as MACD, RSI, and Bollinger Bands to analyze the stock data

- ☐ Analyze the correlation between different stocks by plotting their daily price changes on the same chart
- ☐ Predict future stock prices using machine learning models such as linear regression or neural networks
- ☐ Implement trading strategies based on technical indicators and machine learning predictions.
- ☐ Analyze news sentiment using natural language processing techniques to predict stock price movements.
- ☐ Implement portfolio optimization techniques to determine the optimal allocation of assets across different stocks.
- ☐ Analyze the impact of macroeconomic factors such as GDP, inflation, and interest rates on stock prices.

